

Roadmap for Edge AI: A Dagstuhl Perspective

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ABSTRACT

Edge AI deploys AI methods and capabilities across edge computing resources. Edge AI adapts to data-driven applications, enhances network and radio access, and allow the creation, optimisation, and deployment of distributed AI/ML pipelines with given quality of experience, trust, security and privacy targets. The edge AI research community investigates novel ML methods and network systems for the edge computing environment, spanning multiple sub-fields of computer science, engineering and ICT. The paper summarises the discussions of the Dagstuhl Seminar 21342.

KEYWORDS

Edge AI, Edge Computing, 5G Beyond, Future Cloud, Roadmap

1 INTRODUCTION

The premise of edge computing is to reduce communication latencies and improve the network throughput for emerging and existing applications [9]. This is achieved by placing certain application-specific computations on data generating devices or on network infrastructure components near devices at the edge of mobile and fixed networks [4]. Current Artificial Intelligence/Machine Learning (AI/ML) methods that underpin many important distributed applications assume the availability of large-scale computational infrastructure [5], for example, a cloud platform with substantial computing and storage resources. To shed light on the rapidly evolving domain at the intersection of edge computing and AI/ML, referred to as **Edge AI**, a recent Dagstuhl Seminar 21342¹ gathered community input from a diverse range of experts. This article summarises the discussions held during and after the seminar to collate thoughts on both the technical and societal demands for applying AI methods within the context of edge computing.

There is no commonly accepted definition of what the edge is, where it resides, and who provides it across research communities². However, there is a general understanding of its characteristics: compared to traditional cloud platforms, edge computing is located much closer to the data source in latency and hop count, yielding

¹<https://www.dagstuhl.de/en/program/calendar/semhp/?seminr=21342>

²We deliberately avoid marketing-driven sub-classes of edge computing such as fog or mist computing.

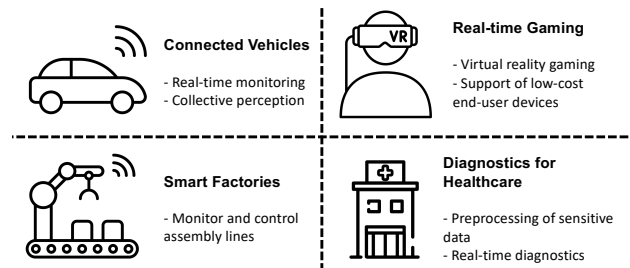


Figure 1: Use cases of Edge AI

increased achievable data transmission rate, but trading this for lower computational power. Each edge node is much smaller than a typical regional cloud data centre and will likely vary significantly in its server count. Compared to end devices and network gateways such as ONTs or cable modems, the edge is assumed to have more computational and storage resources.

A raising area of tension arises from current Artificial Intelligence (AI) and Machine Learning (ML) methods, which require powerful computational infrastructure [5] – a demand that is better satisfied in a homogeneous cloud with ample available computing and data storage resources. However, sending the necessary raw data to the cloud puts pressure on the network w.r.t. bandwidth and throughput. Meanwhile, organisations are usually less keen on sharing (potentially restricted) data with commercial cloud providers. This tension is addressed by the fast evolving domain of Edge AI.

As shown in Figure 1, Edge AI is gradually finding its way to mainstream service domains such as connected vehicles, real-time gaming, smart factories, and healthcare. From an infrastructure perspective, edge environments provide a unique layer for AI by offering a home for embedded AI or federated learning that minimises memory consumption on individual devices, improves privacy by processing data locally on devices and reduces communication needs between distributed entities. These underpin the use cases shown in Figure 1.

This paper aims to present a collective vision that can bring together key actors and enablers to advance the domain of Edge

AI. The discussed perspectives are beyond 5G, future clouds, and AI/ML; further, an envisioned road map and outlook are included.

2 5G BEYOND PERSPECTIVE

The evolution of the fifth-generation (5G) networks towards the 6G era shapes the perspective of future networking. Research considerations here include network communication (e.g. speed, coverage, and resilience) but also quality and delay in computations. To unlock the true potential of complex future networks, various technologies, at both hardware and software levels, need to coexist and cooperate. These include, for example, the creation of edge computing and communication fabric or using self-learning technologies for dynamic network orchestration. There are also critical perspectives in holistic system trustworthiness, including security assurance mechanisms or confidential communication, computing, and learning. In this section, we consider a few relevant challenges and opportunities.

2.1 Human-in-the-Loop Communication and Computation

The computation, communication, and wide-range sensing capabilities available on smartphones and personal carry-on devices are valuable for solving challenges in networking. Examples include local data acquisition as in federated learning, reducing the communication overhead as in device-to-device caching, and cooperating in executing computationally-intensive tasks. In future networks, mobile devices can decide if, when, where, and what fraction of a specific task to offload to a server at the cloud or edge. In addition, they can take on the role of workers, form a pool of compute resources, and divide the tasks based on their preferences to optimise their utility and performance.

While humans or human-driven devices may significantly contribute to Edge AI, their involvement poses several challenges. Such challenges can be modelled using multi-agent systems and addressed by using mathematical tools, such as control and game theory. Furthermore, AI/ML play significant roles if there is some uncertainty and lack of information. In all of the above-mentioned steps, the specific characteristics of humans should be taken into account [1]. For example, humans often act based on heuristics and irrational influences by taking into account factors such as social norms and peer pressure. When dealing with self-interested entities, it is essential to consider mutual trust and to respect the welfare of each entity [11]. Moreover, using humans as a data source, such as using body sensors or GPS will exacerbate privacy concerns that are tightly coupled with legal and ethical challenges.

2.2 Critical but Conflicting Actors and Applications

As we transition from current systems and networks to beyond 5G, the future needs of society will become the driving force that creates new use cases. Innovative technology will be required to address society-driven use-cases [3]. This to an extent is opposed to some of the requirements of current use-cases (such as low-latency and reliability) that are based on technological advances rather than from societal needs. Examples include the current vertical trends, such as resource-efficient manufacturing, green energy generation

and distribution, organic agriculture, and optimization of retail logistics.

Heterogeneous actors build and consequently share edge computing infrastructure to serve their wide range of demands. Despite having overlapping goals, such actors often exhibit conflicting interests; i.e., more benefit for one actor reduces that of the other (here the utility can correspond to higher monetary return), sustainability, improved environmental factors, and the like. Finding a Pareto-optimal and stable solution to this problem is significantly challenging as different utility measures are often conflicting. The problem is aggravated in procedure as it involves several decision-makers instead of a single central authority. That is not only because of self-interest but also due to information asymmetry and different types/preferences [2]. AI can be a solution to this problem, as it enables distributed systems to interact, learn, and make decisions – rendering smart systems inseparable blocks of edge intelligence.

2.3 Edge Intelligence and the Emerging Technologies in Beyond 5G Networks

Next-generation networks beyond 5G encompass several technologies whose efficient deployment depends strongly on reliable cloud infrastructure as well as on edge intelligence. These include, among others, joint communication and sensing, campus networks, open RAN, intelligent reflecting surfaces (IRS), to name just a few. For example, the technology of joint communication and radar sensing is implementable in two networking architectures, namely small cell networks, and cloud RAN [7]. While the latter is amenable to the cloud infrastructure, both implementations greatly benefit from edge intelligence. That is because joint communication and sensing necessitates swift signal processing and precise pattern recognition, both of which are computationally complex.

Another example is campus networks, which covers a geographically limited region to cover the communication requirements specific to that area. For example, a manufacturing company can integrate a campus network in response to the need for secure, reliable, and persistent industrial communications with ultra-low latency. Other applications of campus networks include agriculture fields, construction sites, and hospitals. The 5G technology, together with the edge computing capacity and AI, are the drivers of campus networks. They enable secure and stable communication, fast and no-failure computation, also reliable and efficient performance, even in the absence of precise information. The reliable performance of several other technologies such as IRS depends on edge intelligence. IRS technology relies on the optimal beam configuration, which might happen repeatedly. As a result, the required low-delay computations can be handled by the edge.

2.4 Technology meets Business

Recently, the discussion around 5G and beyond has been largely driven by the potential use cases and the over-arching goal to build a real-time integrated edge computing, AI, and communication service that responds to the dynamic needs of the applications. These anticipated solutions, from everyday life to smart traffic and medical advantages, are immersive but need evaluations within the context of technology and novel business models.

We identified three examples where AI technologies have a role in entirely new functions, however, the business cases and models are yet to be defined for them:

- (1) **Interpreting the results of joint sensing and communication capabilities in future networks.** Future higher frequency communications allow some level of “radar-like” recognition of the environment. This represents a significant change to what current networks do or how they operate, and the security issues involved are substantial.
- (2) **Discovery of optimal link-level communication details through machine learning.** This, in theory, is possible. However, the usefulness of these technologies is still questionable along with the necessary learning costs offset by the optimisation benefits accrued.
- (3) **Interoperability and collaborative use of data and AI technologies.** Current systems are largely run within single organisations, but we raise the question: “what can we do to enable more sharing technologies and inter-operable interfaces”. There is a need for various stakeholders to identify and discuss relevant security, privacy, and ethical issues and tools to respond to them in trustworthy manner.

3 FUTURE CLOUD COMPUTING PERSPECTIVE

The perspective of future cloud computing is shaped by the collaboration between future clouds and future edge nodes. The main reasons for shifting tasks from the cloud to the edge are latency, bandwidth, locality of data, scalability, accessibility, and fault tolerance [4]. This convergence is not only challenging technically, but also involves business and stakeholder challenges. Such an important challenge is the potential competition between the cloud and edge providers. They will need to work together closely to provide the best service offerings to the customers. The technical challenges that form the cloud perspective are considered, including: i) resource management, ii) energy constraints and efficiency, iii) security, trust and privacy, and iv) intermittent connectivity.

3.1 Resource Management

Although edge computing intends to install a large number of resources near users, the necessary locality inherently limits the number of *available* edge resources to far cloud drastically. Such limitations in resource availability may restrict the number and types of machine learning tasks that can be executed at the edge. While the number of such tasks at the edge can be increased through on-demand loading and execution of the corresponding models, this may correspondingly increase the latency of tasks that need to be loaded. Therefore, for latency-critical tasks, reserving resources might be an absolute necessity. As the reservation of resources could drastically limit the number of supported machine learning tasks, the necessity of reserving resources should be determined based on the risk of failure together with the consequences of that failure. Furthermore, the management of these resources in critical situations might be challenging in environments with sparse resource availability as retroactive reservation may result in denial of service.

Importantly, AI tasks at the edge are often embedded in larger settings, e.g., as part of a control workflow. As a result, different component services (belonging to the same application) hosted on edge resources have to interact with each other. Not taking into account that data items may have to be forwarded to services may lead to the fact that these services are not ready when receiving data and/or do not possess the computational resources for handling data items. Accordingly, a backpressure of data items might occur. This is especially the case in stream processing scenarios, e.g., when new data items need to be classified and, based on the classification, forwarded to different destination services. Thus, it might be interesting to observe and analyze the execution of tasks depending on their priority level given different loads on the edge.

3.2 Energy Constraints and Energy Efficiency

In general, edge nodes are expected to be less efficient (in terms of energy and cost) than cloud data centers. That is, as the economies of Scale might work against edge data centers, e.g., by allowing for better cooling, reduction in maintenance due to centralized operations, etc. Both, edge nodes and cloud data centers, have the possibility to deploy renewable energy generators, however the energy efficiency outcome of such approaches are unclear. On the one hand, large cloud centers provide more opportunity to deploy renewable energy generators, increasing their sustainability and environment compatibility, but consuming a significantly higher amount of energy compared to edge nodes. On the other hand, edge nodes might harvest their own energy more easily than cloud data centers due to their small scale and geographical dispersion, even though their capacity for renewable energy units is quite limited.

Additionally, the data need to be transferred via the Internet which also consumes energy – increasing the potential of edge nodes to surpass energy-efficient cloud data centers. Edge nodes can pre-process data, e.g., can detect anomaly cases and transmit only relevant data or adapt sensing rates dynamically. However, energy saving largely depends on the amount of processing that can be done at the edge. Another possibility is to move the training process partially to the edge, utilizing methods such as Federated Learning (see Sec. 4.3). In that case, only learned parameters (e.g., gradients) need to be transmitted, rather than raw data, which is multiple orders of magnitude larger. Such reduction in network usage due to edge can translate to high energy saving for bandwidth-hungry applications, such as real-time video processing, since live camera feeds can be resolved by ML models deployed closer to the users.

Another potential advantage of the edge nodes is the more specially designed hardware compared to generic cloud server racks, as this specially designed hardware is becoming more popular in the market (hardware accelerators, NCS2, etc.) and utilizes limited processing and graphical capacities more efficiently.

3.3 Security, Trust, and Privacy

As shown in Figure 2, edge AI offers both opportunities for higher trust, security, and privacy, but also adds additional challenges. Security might be increased as some attacks might be detected early and countermeasures can be taken. However, the number of

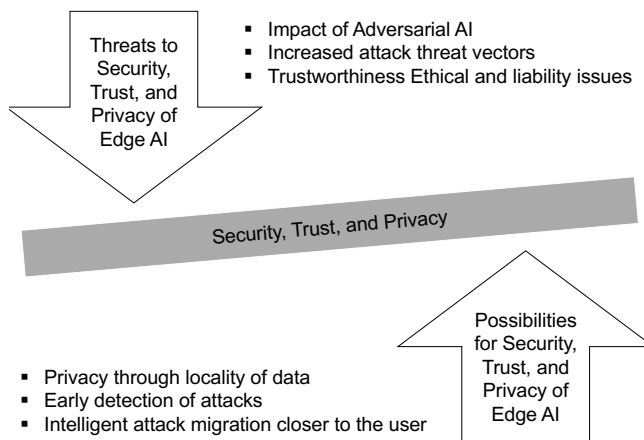


Figure 2: Trust, Security and Privacy of Edge AI

possible targets is much higher, which leads to a higher potential of attacks.

Edge AI might have an issue with trust in the system, which is caused by its distributed nature and the potential liability issues. However, the trust in edge servers might be increased through open implementation and specifications. Additionally, it might be necessary to differentiate between devices under the control of the user and resources provided by providers (e.g., AWS, GCP, Azure). One possible solution for increasing trust is the introduction of a reputation system known from other areas, that assesses and manages trust. However, it will take new players a while to build that reputation.

Another property of edge AI is that the data required to perform the machine learning tasks is kept local at the edge servers. This may lead to an assumption that the privacy and trust in edge servers is higher compared to the cloud. However, the edge by itself does not guarantee privacy, but raises new challenges to ensure privacy. That is, as the trust of edge nodes is harder to keep and manage compared to a cloud provider, thus also aggravating data security and privacy, even though the locality of the data will prevent some attacks on the data.

3.4 Intermittent Connectivity

In general, we also consider the possibility of poorly connected edge servers, i.e., edge servers in regions where the connection to the Internet is poor. Also in those areas, edge intelligence has a huge potential and can be a driver towards digitalized non-urban areas. Some examples for such systems include water and air pollution monitoring and natural disaster (wildfire, flood, volcanic eruption, etc.) prediction. For instance, in water pollution monitoring, the identification of substances in the water could be done using edge resources.

The lack of reliable network connectivity in those areas could be the main driving factor in environmental monitoring scenarios in which streaming data has to be processed in near real-time. Intermittent connectivity and energy constraints prevent monitoring systems from continuously transmitting raw data to the cloud for processing, whereas less data-intensive control signals like the

output of ML algorithms can still be communicated under interruptions and low bandwidth availability. In addition, the actors which rely on the outcome of ML algorithms are often close to the data sources. By not sending the data to the cloud, a complete processing step can be saved, e.g., starting countermeasures autarkically. In the previous example of water pollution monitoring, the services running at the edge can then inform pumps or valves to open or close, depending on the scenario.

4 EVOLVING AI/ML PERSPECTIVE

In this section, we describe the AI perspective on edge-nodes and the challenges associated with it. Figure 3 depicts the different complexity of AI and the availability of data on mobile devices, edge servers, and cloud servers: While the mobile devices have the highest volume of data available to them, their capability of processing this data is generally limited. Thus, preprocessed data can be offloaded to local edge servers to perform tasks which cannot be executed by the mobile devices. The edge server has more resources than the mobile devices and can run more complex tasks, but due to the preprocessing its access to data is more limited. Tasks that cannot be executed will then again be offloaded to the cloud server, which can run the most complex models, but also receives only a part of the data available at the edge server.

In the following, we describe three challenges that have been identified and deserve community attention: These challenges are 1) availability of accelerators for AI applications; 2) defining a trade-off between accuracy and resource demand; and 3) utilising federated learning of edge servers.

4.1 Accelerators for AI Applications

While we expect that hardware accelerators for AI applications will also be available at the edge, it is expected that only small hardware accelerators will be available there. That is, as resources at the edge can be used less elastically compared to the cloud, which might lead to idle hardware. Due to business aspects, it is expected that the large hardware accelerators will be deployed in the cloud.

One current trend to work with these limited resources is to split the model and AI algorithm into multiple smaller chunks, which can be executed by these small hardware accelerators. Through joint interference among these small hardware accelerators, the user experience and quality of service of applications running at the edge can be enhanced. In general, it will be necessary to transfer the large models with high resource demands to the cloud and vice-versa.

Machine learning also provides unique opportunities for off-loading and multi-tier architectures. Unlike traditional workloads, which inherently offer conflicting requirements in terms of scaling to more nodes and tiers and keeping the state of the workload sufficiently consistent between the different nodes, machine learning, especially training of neural networks, has clearly defined state distribution and synchronisation models [10]. Furthermore, unlike traditional distributed systems the consistency requirements can be comparably easily relaxed, thereby trading network bandwidth and number of replicas for typically moderate losses in accuracy. An important question is therefore which accuracy and latency

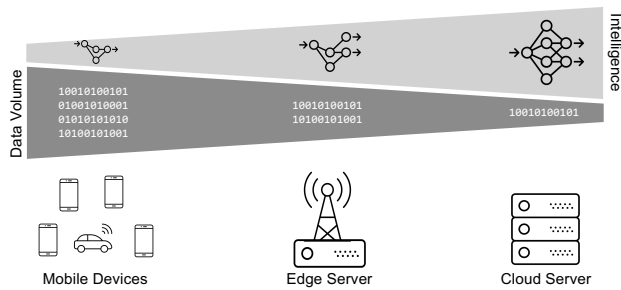


Figure 3: Scope of Edge AI

requirements the application has and how to monitor the quality of the model during the entire life-cycle on the edge.

4.2 Tradeoff Between Accuracy and Resource Demand

As mentioned previously, edge servers have higher resource constraints compared to cloud servers. Some models may only run in the cloud due to their complexity.

In today's AI research, the common goal is often to achieve results with the highest possible accuracy or maximizing the reward function. However, a pivotal difference for edge AI is that the quality of the result is not the only measure for performance, but also metrics such as energy consumption and memory footprint. Thus, models need to be elastic to adapt to the currently available resources and memory, even at the costs of accuracy. For that purpose, it must be possible to adapt the size of the respective machine learning models. This can be done by miniaturizing them using techniques like transprecision, quantization or approximation. These techniques generally provide a tradeoff between performance metrics and the accuracy of the model. Examples for Edge AI minimization are TinyML and TensorLite. An interesting research challenge is the analysis of adapting the machine learning model given the current (and maybe future) available resources. In addition, the potential of offloading partitioned models to the cloud has been explored. If the accuracy of a miniaturized model executed at the edge is insufficient, then offloading certain layers of the models will alleviate the computational burden on the edge. The cloud can then be able to provide the results with high accuracy and send it back to the edge.

4.3 Federated Learning

In addition to the miniaturization of models, models can be trained on multiple devices simultaneously using federated learning. This is especially important for edge AI, as the distributed nature of edge-servers increases the need of federated learning.

For federated learning, the data is kept local and thus not shared with the cloud. However, when a complex model is run at an edge-server, the available resources might be insufficient to run or train the model. One research challenge is the deployment of a complex model only in the cloud, which is trained in a federated manner based on less complex models deployed at the edge.

Federated learning also allows for the classification of time-critical labels at the edge directly, while other labels need to be

classified in the cloud. This adds another layer of elasticity to the model. We expect that the accuracy of these time-critical labels remains high even though the model might be simplified through miniaturization by summarising all non-time-critical labels to be classified by the cloud model. Thus, open research topics are the miniaturization of a model such that the accuracy for a subset of labels remains high and the decision on when a classification needs to be executed by the cloud.

In addition, the data received by the edge servers and generated by the mobile devices might be quite diverse. Federated learning can help to train a single model using these diverse sensor inputs and increase the accuracy of all models independent of their sensor setup. The detection of malicious data, that could harm the training process, is also considered as an open challenge.

Another challenge is the elastic adaption of federated learning systems to dynamically updated client partnerships considering the satisfaction of collaboration criteria (e.g., minimum number of training partners) and still optimising the model accuracy.

5 ROADMAP AND OUTLOOK

Roadmap overview: A roadmap envisioned by the community for Edge AI is highlighted in Figure 4. Along with three identified driving areas of 5G and beyond, future cloud, and evolved AI/ML, the advancement of different technologies and the growing business interests will take Edge AI forward in terms of hardware, software, service models, and data governance.

Starting from the current state of play driven by cellular, cloud, and AI/ML service providers, the roadmap reflects five general phases: a scalable framework, trustworthy co-design, sustainable and energy-efficient deployment, equal accessibility, and pervasive intelligent infrastructure. As changes always occur, the order of the items in the sequence presented in the roadmap may be changed or the items combined. Nonetheless, this Edge AI roadmap reflects the combined effects of technology enablers and non-tech demands, such as socioeconomic transformation of user behaviours, purchasing power and business interests.

Open research challenges: Despite its promise and potential, Edge AI faces challenges ahead in large-scale deployment, including energy optimization, trustworthiness, security, privacy and ethical issues.

As an important goal of sustainability, the energy consumption of Edge AI needs to be optimized. Energy efficiency is crucial for embedded infrastructure processing Edge AI (e.g., road side units, micro base stations) to sustainably support advanced autonomous driving and Extended Reality (XR) services. Through the pipeline of data acquisition, transfer, computation, and storage, there exists the possibility for Edge AI to trade off accuracy to less power and less time consumed. For instance, noisy inputs from a wide-range of sensors can be selectively processed and transferred in order to save energy. A set of applications would be satisfied with an 'acceptable' accuracy instead of exact and absolutely correct results. By introducing this new dimension of accuracy to the optimization design, the energy efficiency can be further improved.

Concerning trustworthiness, Edge AI benefits from its close proximity to end-devices and users. However, due to the distributed deployment and deep penetration into personal contexts, the safety

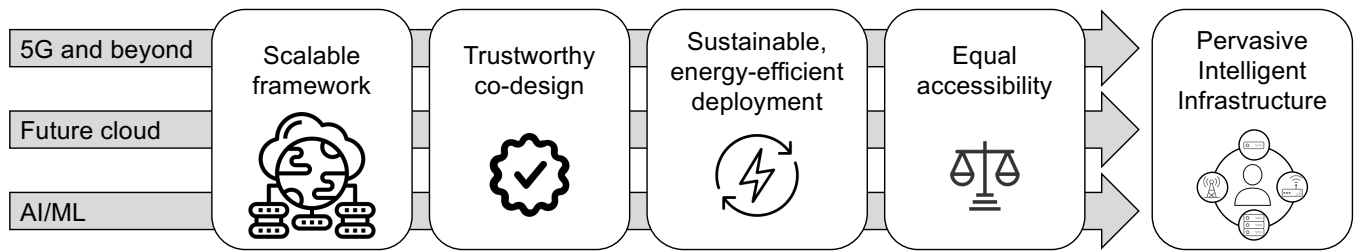


Figure 4: Roadmap of Edge AI

and perceived trustworthiness for Edge AI services are raising concerns among the stakeholders (for example, end users, public sectors, ISP) [8]. To achieve trustworthy Edge AI, critical building blocks are needed, including verification and validation software-hardware mechanisms that ensure transparency and explainability in the training and deployment of Edge AI in decentralized, uncontrolled environments. The trustworthiness of Edge AI is a stepping stone to establish an appropriate governance and regulatory framework, on which the promise of Edge AI can be built.

Security and Privacy/Ethical Issues: There are two ways to consider the security and privacy aspects of Edge AI: 1) usage of Edge AI to provide security and privacy; 2) new security and privacy issues due to the use of edge AI (Figure 2). Edge AI can be a vital tool to ensure security and privacy within a network [6]. Instead of performing AI processing in the Cloud, Edge AI processes the data locally closer to the user. That prevents the necessity of transferring data between the user and the Cloud and reduces the possibility of attacks during data transmission in the backhaul.

Moreover, privacy can be improved by keeping and processing data in the local vicinity. Processing is focused on moving the interface of the AI workflow to the device while maintaining data restricted to the device. Another security benefit of edge AI is the possibility of an AI algorithm decentralized and eliminating a single point of failure in the cloud AI system. In a way, Edge AI could be a means to limit the impact of an attack on the local environment and mitigate it already at the edge level due to added intelligence. Yet, new security and privacy issues can arise with the use of edge AI. Decentralisation of Edge AI opens up unknown attack vectors and increases the number of entry points for AI system attackers. Edge AI devices might not have the same level of security as in the Cloud, which can be used as easy entry points to get access to attack the AI system. Moreover, Edge devices are physically accessible, and adding AI to edge devices makes the impact of captured edge devices severer than in case of an edge device without AI. For instance, taking control of Edge AI devices might jeopardise or take control of almost all the localised network services of a particular location.

Concluding remarks: The promise of Edge AI comes with new challenges and uncertainties. This paper is our endeavour to capture the latest technological development, crucial actors across three major dimensions of 5G-beyond, future cloud and AI/ML for the envisioned roadmap. The authors anticipate that the perspective presented in this article may provide a helpful view of Edge AI to the community and further pave the way for a global rollout.

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